Learning Through the Lens of System Change:

A Sociocultural Analysis of Teamcenter NEXPHASE PLM Migration Training at SilicaLayer Technologies

Introduction

SilicaLayer Technologies is a global semiconductor equipment company that recently migrated from IntegraPLM to Teamcenter as its primary product NexPhase PLM platform. This migration came with restructured workflows, new roles, and new responsibilities for cross-functional teams — engineers, supply chain analysts, program managers, and quality professionals. To support this transition, our instructional design team built a training ecosystem that included day-in-the-life (DITL) eLearning courses, work instruction documents, instructor-led training (ILT) sessions, and various job aids. I write this analysis as both the instructional designer who built parts of this program and as someone embedded in the organization who has watched how employees actually experience it. That dual perspective is my observational lens for this paper.

Part 1: Fifteen Concepts in Practice

1. ***Acquisition Metaphor*** *(Sfard, Week 2)*

The Teamcenter training program is built almost entirely on the acquisition metaphor — knowledge is packaged into modules, delivered to employees, and verified through completion tracking and knowledge checks. When an employee's progress bar reaches 100%, the organization considers them trained and ready to work in the live system. In practice, I have seen engineers who completed every module struggle significantly the first time they opened Teamcenter on a real production day, because the module environment and real work are very different contexts. The metric looks clean, but what it measures is completion, not capability.

2. *Participation Metaphor* *(Sfard, Week 2)*

The participation metaphor — learning by becoming part of a community through real doing — is almost entirely absent from the formal training program. There is no structured path through which a new Teamcenter user moves from beginner to confident practitioner alongside experienced colleagues. The ILT sessions offer brief shared space, but the structure remains one-directional. Employees finish training having acquired information about Teamcenter without yet feeling like they belong to the Teamcenter-using community at SilicaLayer Technologies.

3. *Human-Centered Design* *(Gutiérrez & Jurow, Week 3; Remake Your Class video)*

Our team made a real effort toward human-centered design by choosing the DITL format — structuring content around what an employee actually does in a real workday rather than around system features in a menu tour. However, the design process was still driven heavily by subject matter experts and project timelines rather than by learners themselves. Feedback from pilot users revealed workflow inaccuracies that reflected how SMEs thought work happened, not how employees actually did it. A more fully human-centered process would have started with sustained observation of real work before any module was built.

4. *Participatory Design* *(Gutiérrez & Jurow, Week 3)*

End-users — the engineers and supply chain coordinators who would live with the new workflows — were consulted in review cycles but were never part of the design team itself. Their role was to validate content that had already been built, not to shape what was built or how. Several engineers flagged during review that task sequences did not match how their teams actually managed engineering change orders, but many of those corrections were deferred due to timeline pressure. A genuinely participatory process would have caught those mismatches before development, not after.

5. *Equity-by-Design* *(Gutiérrez & Jurow, Week 3)*

The training was designed with an implied default learner: English-fluent, digitally comfortable, NEXPHASE PLM-experienced, with protected time to complete training before go-live. Not all SilicaLayer Technologies employees match that profile. Some team members were navigating Teamcenter training while managing peak production demands; others had limited prior enterprise system experience; some were more comfortable in languages other than English. The materials were not differentiated to support these learners, which meant the program worked best for those who needed the least help — a quiet but real inequity.

6. *Third Space* *(Gutiérrez & Jurow, Week 3; Lave & Wenger, Week 7)*

Every employee in this training program already knew how to do their job — they just did not yet know how to do it in Teamcenter. That gap between deep work knowledge and new system knowledge is exactly where a third space could have been created. Instead, the modules presented the Teamcenter way of doing things as complete and authoritative, with no deliberate bridging to what employees already knew from NexPhase PLM . A simple design move — acknowledging what changed, what stayed the same, and how prior knowledge connects to the new system — would have opened that third space and made learning feel additive rather than replacive.

7. ***Identity in Learning*** *(Week 4 — Sean Numbers & Kindergarten Physics videos)*

The migration repositioned experienced, NexPhase PLM-fluent professionals as beginners, and the training program did nothing to acknowledge or soften that shift. Senior engineers who had managed complex engineering change processes in NexPhase PLM for over a decade sat through demonstrations of basic NexPhase PLM concepts alongside brand-new hires. In ILT sessions, I observed some of these senior staff disengaging — not because the content was too hard, but because the training treated them as though their years of expertise were invisible. Identity in learning matters because when people do not feel recognized, they stop participating.

8. ***Funds of Knowledge*** *(Week 4 readings)*

The employees moving through Teamcenter training are remarkably knowledge-rich — supply chain analysts understand global sourcing dynamics at SilicaLayer Technologies, quality engineers understand documentation compliance stakes, program managers understand cross-functional dependencies. None of this was treated as a resource in the training design. Modules opened from the system, not from the person, and never asked what employees already knew before showing them something new. Several employees told me they felt like they were learning a new tool in a vacuum, disconnected from the actual work they had been doing for years.

9. *Discourse & Participation Structures* *(Week 4 — Sean Numbers video; Week 5)*

The dominant participation structure across both the eLearning and ILT components is one-directional: the training speaks, the learner receives. In the DITL courses, the only moment of reversal is the knowledge check, where learners select a predetermined correct answer. In ILT, participants were largely on mute in virtual sessions, with the instructor demonstrating while chat ran in the background. There was no design space for employees to share their NexPhase PLM experience, raise concerns about workflow changes, or problem-solve collectively — the very kinds of exchanges that would most accelerate real learning in this context.

10. *Turn-Taking & Participation Patterns* *(Week 5 — Transcript Explorer; classroom interaction analysis)*

If we were to visualize the turn-taking pattern of a Teamcenter DITL module, we would see a single unbroken arrow from module to learner — the module narrates, the learner clicks Next, repeat. The only learner "turn" is a knowledge check with no genuine response channel. Even in ILT, the participation pattern was heavily skewed toward the instructor; questions from participants were typically short and procedural rather than substantive. This stands in stark contrast to the Sean Numbers classroom, where student thinking drove the conversation and the teacher's role was facilitative rather than transmissive.

11. ***Multimodality*** *(Week 5 — Kindergarten Physics video; Embodying STEM readings, Week 8)*

The training materials engaged multiple modes more than typical corporate eLearning — DITL simulations let learners click through real Teamcenter interface screens, and work instruction documents paired written steps with annotated screenshots. However, there was no embodied practice with real data in a live Teamcenter environment, and no opportunity to manipulate actual engineering artifacts. The kindergarten physics video showed how powerfully physical, object-based engagement supports learning; the Teamcenter training never created anything close to that kind of hands-on contact with real system work.

12. ***Lifelong & Lifewide Learning*** *(Week 6 — Lifelong/Lifewide graphic; Mimi Ito)*

The training program was designed as though it were the beginning of each learner's relationship with NEXPHASE PLM systems and with SilicaLayer Technologies's processes. In reality, many employees had used SAP or other enterprise systems at previous companies, had self-taught workarounds in NexPhase PLM, or had built deep informal knowledge of how product data flows in the semiconductor industry. There was no intake process, no differentiated pathway for experienced users, and no design acknowledgment that most real learning in a new system happens outside formal training. The life wide dimension of these learners was simply not part of the design picture.

13. ***Interest-Driven / Connected Learning*** *(Week 6 — Mimi Ito; Connected Learning Alliance)*

Most employees were not interested in learning Teamcenter as a system — they were interested in being able to do their jobs accurately and without delays. An engineer's real interest is moving an engineering change through approval without holding up the production line; a supply chain analyst's real interest is ensuring parts availability without a missed delivery. The training was designed around system functionality, not around these underlying professional goals. The DITL format was a step toward relevance, but the scenarios were often too clean and linear to connect to what employees actually cared about in their daily work.

14. *Informal Learning Environments* *(Week 6 — Mimi Ito; informal learning readings)*

The most active learning ecosystem around the Teamcenter migration has been informal — employees created their own Teams channels for questions, early-confident users became unofficial go-to resources, and some teams built their own quick-reference guides that have become more widely used than the official job aids. I only learned about some of these resources through colleagues mentioning them in passing. The formal training program treats itself as the complete solution, while the real learning is happening in the spaces around it — unacknowledged, unsupported, and unmeasured.

15. *Situated Learning* *(Week 7 — Jean Lave & Etienne Wenger; Teaching as Learning in Practice)*

Situated learning holds that knowledge is inseparable from the context, activity, and relationships in which it is learned — and this is the most fundamental critique I can make of the Teamcenter training program. Employees learned about Teamcenter in a simulation environment that carried no real consequences, contained no real SilicaLayer Technologies data, and demanded none of the judgment that real production work requires. The employees who became most capable in Teamcenter were those who had early access to the live system alongside someone more experienced doing real work — not those who completed the most modules. That is situated learning, and the formal program did not design for it.

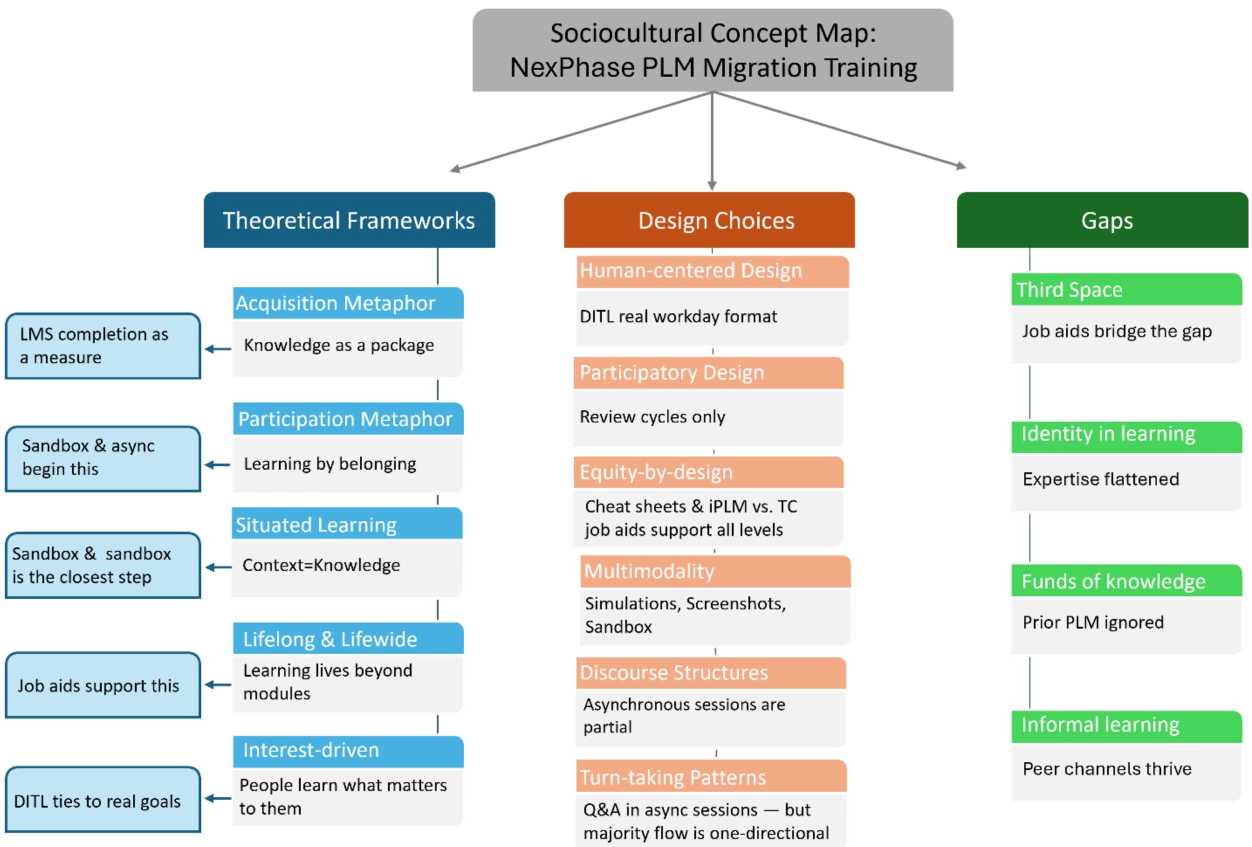
Summary: Fifteen Concepts at a Glance

The table below summarizes the presence or absence of each concept and its consequence in the Teamcenter training environment.

Concept	Status in Training	Consequence
1. Acquisition Metaphor	Dominant — drives module design	Masks real learning gaps
2. Participation Metaphor	Minimal — brief ILT moments	No community entry path
3. Human-Centered Design	Partial — DITL format attempt	SME-driven, not learner-driven
4. Participatory Design	Absent — users validate only	Misaligned content sequences
5. Equity-by-Design	Not addressed	Excludes non-standard learners
6. Third Space	Not created	NexPhase PLM knowledge never bridged
7. Identity in Learning	Flattened — all treated as novices	Senior staff disengage
8. Funds of Knowledge	Ignored	Experts feel dismissed
9. Discourse & Participation	One-directional	No learner voice in design
10. Turn-Taking Patterns	Instructor-only flow	Passive ILT + click-through eLearning
11. Multimodality	Partial — sim + screenshots	No live system practice
12. Lifelong & Lifewide	Not acknowledged	Prior experience invisible
13. Interest-Driven Learning	System-focused, not goal-focused	Low motivation, poor transfer
14. Informal Learning	Active but unrecognized	Peer help exceeds formal training
15. Situated Learning	Violated by simulation design	Real work context missing

Part 2: Concept Map

The concept map on the following page organizes all fifteen concepts around a single central tension: the gap between the acquisition-dominant training program that currently exists and the participation-based, identity-aware, situated learning environment that sociocultural theory calls for. Concepts are grouped into three clusters — how learning is theorized, what is present in the design, and what is missing. The arrows show relationships and tensions between clusters.



Reflection

Writing this analysis has clarified something I have sensed but not fully named in my work as an instructional designer at SilicaLayer Technologies: the Teamcenter training program is technically complete but sociocultural thin. It covers the content, tracks completion, and gives the organization a record of readiness — but it does not create the conditions for people to actually become confident Teamcenter practitioners within SilicaLayer Technologies's community of practice.

What these course concepts have given me is vocabulary for the gap. When I watch a senior engineer click through a beginner-level module as if her fifteen years of NEXPHASE PLM experience do not exist, I am watching identity in learning being flattened and funds of knowledge being ignored. When I design a module that opens from the system rather than from the person, I am building a third space that never opens. When employees learn more from a Teams conversation with a colleague than from a forty-five-minute ILT session, I am watching situated learning, and informal learning environments do the work the formal program was meant to do.

None of this means the current design is a failure — it was built under real constraints of time, budget, and organizational expectations. But these concepts ask better questions at the start of a design process: Who are these learners, and what do they already know? What community are they joining, and on what terms? How close can we bring learning to reality? Asking those questions before the first module is built would produce a fundamentally different and more effective training ecosystem for Teamcenter at SilicaLayer Technologies.